

Introduction to Modeling for Formal Verification

Today's goals

1

Module 1

Introduction to Model Checking

Motivation of Verification

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Course Updates and Accessibility Support Reading • 1 min

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Non-Credit Students: Welcome and Where to Find Help Reading • 10 min

✓

Motivating Examples Video • 7 min

✓

High Profile Previous Bugs and Their Impacts Video • 10 min

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Principles of Model Checking - Chapter 1, Introduction - Part 1 Reading • 15 min

Hardware and Software Bugs - Wikipedia Reading • 20 min

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Motivation of Verification Quiz Graded Assignment • 10 min

Overview of Verification

Overview of Functional Verification Video • 5 min

Techniques for Functional Verification Video • 10 min

Functional Verification Reading • 30 min

Principles of Model Checking - Chapter 1, Introduction - Part 2 Reading • 15 min

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Overview of Verification Quiz Graded Assignment • 10 min

Formal Verification and Model Checking

Overview of Formal Verification and Model Checking Video • 7 min

Elements of Model Checking Video • 6 min

Challenge to Model Checking: State Space Explosion Video • 5 min

Principles of Model Checking - Section 1.1, 1.2 Reading • 30 min

Formal Verification and Model Checking - Wikipedia Reading • 30 min

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Formal Verification and Model Checking Quiz Graded Assignment • 10 min

Module 2

Modeling

Module 3

Modeling System Composition

Principles of Model Checking - Chapter 1, Introduction - Part 1

0:07 / 0:37

2x

Principles of Model Checking – Chapter 1 (page 1 – 6)

This part introduces verification for hardware and software systems, and indicates their importance and practical impacts.

Accessing the Textbook

The textbook is a supplemental reading.

For CU affiliates, you can access the textbook at this link: [Principles of Model Checking](#) (external link; opens in a new tab).

For non CU affiliates, you can access the textbook through your institution.

Christel Baier and Joost-Pieter Katoen. *Principles of Model Checking*, MIT Press, 2008. ISBN 978-0-262-02649-9.

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